

Trained

Large labelled
training set

train

Corpus-specialized
features

Standard

Cross-env.



Large labelled training set

Standard



train



Corpus-specialized features

Cross-env.



Training-free, prior work

Frozen patch features

Codebook or Segmentation CLS attention or

Prior-selected / aggregated features

Standard **×** Cross-env. **×**



Frozen patch features

Standard



Codebook

Prior-selected / aggregated features



Segmentation



Cross-env.



Ours

Frozen patch features → Within-image
Balanced, matchable patch evidence

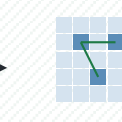
Across-image

Standard ✗ Cross-env. ✗



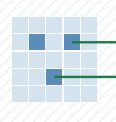
Frozen patch features

Standard



Within-image

Balanced, matchable patch evidence



Across-image

Cross-env.



Training cost vs. performance

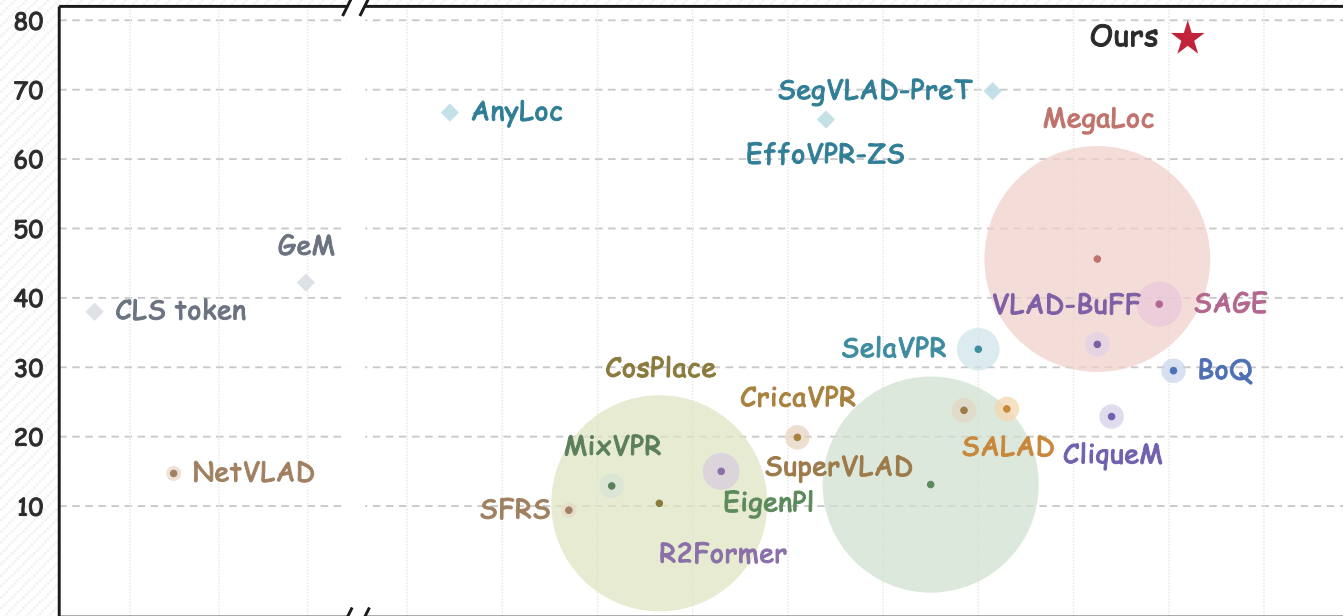
VP-Air R@1 (cross-environment benchmark, %)

Tokyo 24/7 R@1 (standard benchmark, %)

★ ours (training-free) ♦ training-free ● trained ● 1 million images ● 5 million images

Method	Tokyo 24/7 R@1 (%)	VP-Air R@1 (%)	Training Cost	Training Data Size
Ours	98.5	78	Training-free	0
AnyLoc	83.5	68	Training-free	0
SegVLAD-PreT	90.5	68	Training-free	0
EffoVPR-ZS	90.5	62	Training-free	0
MegaLoc	96.5	65	Trained	5 million
VLAD-BuFF	97.5	40	Trained	5 million
SAGE	98.5	40	Trained	5 million
BoQ	98.5	30	Trained	5 million
SelaVPR	94.5	33	Trained	1 million
CosPlace	87.5	30	Trained	5 million
CricaVPR	90.5	25	Trained	1 million
SALAD	94.5	24	Trained	1 million
NetVLAD	58.5	15	Training-free	0
SFRS	85.5	10	Trained	1 million
MixVPR	86.5	13	Trained	1 million
EigenPI	88.5	10	Trained	1 million
R2Former	88.5	15	Trained	5 million
SuperVLAD	90.5	18	Trained	1 million
CLT token	52.5	38	Training-free	0
GeM	70.5	43	Training-free	0
CLIQUE	97.5	23	Trained	1 million

VP-Air R@1 (cross-environment benchmark, %)



Tokyo 24/7 R@1 (standard benchmark, %)

★ ours (training-free)

training-free

- **trained**

● 1 million images

5 million images

Performance across 17 benchmarks

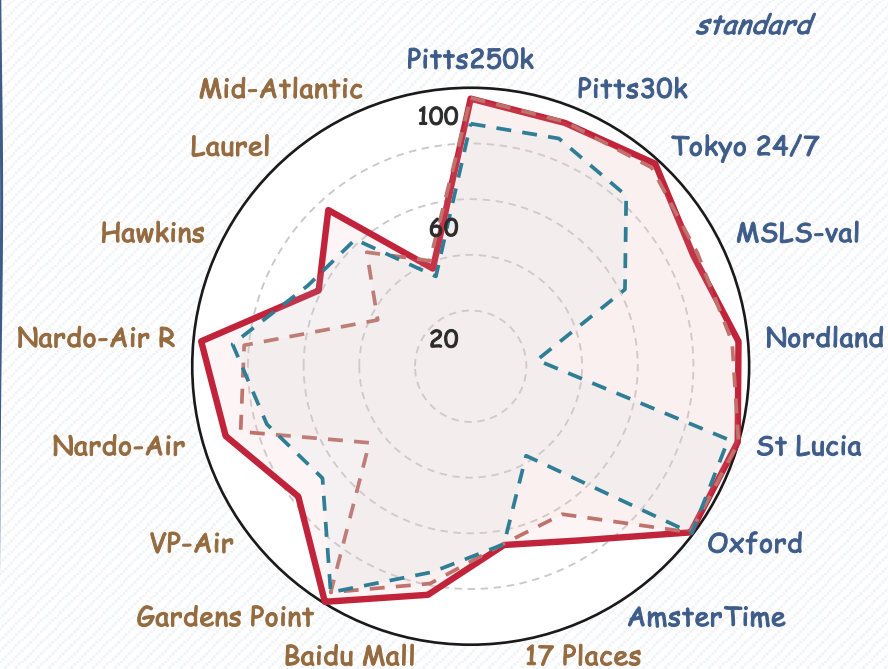
standard

Pitts250k, Pitts30k, Tokyo 24/7, MSLS-val, Nordland, St Lucia, Oxford, AmsterTime, 17 Places, Baidu Mall, Gardens Point, VP-Air, Nardo-Air, Nardo-Air R, Hawkins, Laurel, Mid-Atlantic

cross-environment

Ours, MegaLoc, AnyLoc

Benchmark	Ours	MegaLoc	AnyLoc
Pitts250k	100	100	100
Pitts30k	100	100	100
Tokyo 24/7	100	100	100
MSLS-val	100	100	100
Nordland	100	100	100
St Lucia	100	100	100
Oxford	100	100	100
AmsterTime	100	100	100
17 Places	100	100	100
Baidu Mall	100	100	100
Gardens Point	100	100	100
VP-Air	100	100	100
Nardo-Air	100	100	100
Nardo-Air R	100	100	100
Hawkins	100	100	100
Laurel	100	100	100
Mid-Atlantic	100	100	100



cross-environment

— Ours

— MegaLoc

AnyLoc